

Pipeline Report: cat000001

Neutrino Momentum Integration

Date: November 19, 2025 | Category: cat000001 | Pipeline: tick3_ch2_min2_tot3_e2p0

Executive Summary

✓ **Successfully integrated neutrino momentum into the complete pipeline**

The pipeline has been updated to include true neutrino momentum in both cluster and volume metadata, enabling ED/MT/CT analyses without ROOT files.

Cluster Images Summary

Plane	Files	Clusters	Main Tracks	MT %	Metadata
U	1	162	38	23.5%	18 cols ✓
V	1	200	38	19.0%	18 cols ✓
X	1	204	38	18.6%	18 cols ✓
TOTAL	3	566	114	20.1%	✓

Volume Images Summary

Plane	Files	Volumes	Shape	Neutrino Mom.
U	1	38	(208, 1242)	✓ Present
V	1	39	(208, 1242)	✓ Present
X	1	39	(208, 1242)	✓ Present
TOTAL	3	116	1m × 1m	✓ Verified

Neutrino Momentum Statistics (X Plane)

Component	Min	Max	Mean	Units
p_x	-0.058	0.000	-0.011	MeV/c
p_y	0.000	0.017	0.003	MeV/c
p_z	0.000	0.044	0.008	MeV/c
Magnitude	0.000	0.075	0.014	MeV/c

Non-zero entries: 73 / 204 (35.8%) — True neutrino interactions

Implementation Status

- ✓ C++ (Clustering.cpp, Cluster.h/cpp): Added true_neutrino_mom branches
- ✓ Python (generate_cluster_arrays.py): Extended metadata 15 → 18 columns
- ✓ Python (create_volumes.py): Added neutrino momentum to volume metadata
- ✓ End-to-end verification: Full pipeline tested on cat000001

TPStream Resubmission Status

Type	Required	Found	Missing	Progress
ES	5960	5758	202	96.6%
CC	50660	46218	4442	91.2%

Conclusions

- ✓ **Neutrino momentum successfully integrated into cluster and volume metadata**
- ✓ **Pipeline verified on cat000001 with correct metadata structure**
- ✓ **Data ready for ED/MT/CT analyses without ROOT file access**
- ✓ **96.6% ES and 91.2% CC tpstream files complete; resubmissions ongoing**

The implementation allows downstream ML models and physics analyses to access true neutrino momentum directly from NPZ files, significantly improving